MILITARY SPECIFICATION

INHIBITOR, CORROSION, LIQUID COOLING SYSTEM (DICHROMATE TYPE)

1. SCOPE

1.1 Scope.- This specification covers an inhibitor suitable for use in closed cooling water systems of internal combustion engines; the inhibitor possesses anti-corrosive properties without ingredients which would be injurious to such cooling systems.

1.2 Classification .-

1.2.1 The inhibitor furnished under this specification shall be of one type and grade as hereinafter specified.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-B-41	Battery Water
O-C-275	Chemicals, Photographic "Single Substance" General Specification for
O-S-595	Sodium Dichromate, Technical Grade (Sodium Bichromate) Sodium Dichromate Dihydrate
PPP-C-96	Cans, Metal 28 Gage and Lighter

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.-

3.1.1 Sodium dichromate.- The sodium dichromate used in the inhibitor shall conform to the requirements of Specification O-S-595.

3.1.2 Sodium hydroxide.- The sodium hydroxide used in the inhibitor shall conform to class II of Specification O-C-275.

3.1.3 Distilled Water - The distilled water used in the inhibitor shall conform to the requirements of Specification O-B-41.

3.2 Composition.- The inhibitor shall consist of a solution of the ingredients listed in Table I; the quantities shown in the table are those required for one gallon of the solution.

TABLE I

Sodium dichromate - $Na_2Cr_2O_7.2H_2O$	46.0 ounces
Sodium hydroxide - NaOH (minimum 98 percent)	12.9 ounces
Distilled water - H_2O	113.9 ounces

3.2.1 Prior to packing in containers, the inhibitor shall be filtered to remove all residue.

3.3 Properties

3.3.1 Alkalinity - The pH value of the solution shall be between 9.0 and 10.5 when tested according to the method described in 4.7.1.

3.3.2 Specific gravity - The specific gravity of the solution shall be between 1.29 and 1.31 when tested according to the method described in 4.7.2

3.3.3 Clarity - The solution shall be clear and free from any sediment or foreign matter.

3.3.4 Sodium dichromate content - The inhibitor shall show a sodium dichromate ($Na_2Cr_2O_7.2H_2O$) content of 25.5 to 27.5 percent by tested according to the method described in 4.7.3.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conforms to prescribed requirements.

4.2 Lot.- For purpose of sampling, inspection, and testing, a lot shall consist of all material offered for delivery at one time, manufactured as one batch.

4.3 Sampling.- Two containers shall be taken by the contractor at random from among the containers comprising the lot for testing as specified in 4.7 through 4.7.3.2.

4.4 Inspection.- Inspection may be made at any time during the entire process of manufacture. The passing as satisfactory of any material or method used in the manufacture of the inhibitor shall not relieve the contractor or responsibility for an unsatisfactory finished product. Final inspection of the finished product shall be made at the point of delivery designated in the contract or purchase order.

4.5 Containers.- Sample filled containers (quart or gallon) selected on the basis of four out of each 100 in the lot shall be examined for defects of construction of the container and for evidence of leakage, for unsatisfactory markings, and all other preparation for delivery requirements. Each filled container shall also be weighed to determine the amount of contents. Any container having one or more defects, or under required fill, shall be rejected and if the number of defective containers exceeds two per each hundred in the lot, the lot shall be rejected.

4.6 Rejection.- Lots shall be accepted or rejected by the Government inspector on the basis of the laboratory test report.

4.7 Tests.-

4.7.1 Alkalinity.- The p^H value of the solution as received shall be determined with suitable p^H meter accurate to 0.1 p^H.

4.7.2 Specific gravity.- The specific gravity of the solution as received shall be determined at 25/15.6°C., by means of a suitable hydrometer accurate to one in the second place.

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4.7.3 Sodium dichromate content.- Transfer to a weighing bottle about 7 ml of the sample received. Weigh the sample and transfer quantitatively the contents of the weighing bottle to a 500 ml. volumetric flask, dilute to mark and mix thoroughly. Transfer a 50 ml. aliquot to a 250 ml. glass-stoppered Erlenmeyer flask, add 4 to 5 grams reagent grade potassium iodide crystals and shake to dissolve. Add 10 ml. diluted sulfuric acid (1:2) and allow to stand 10 minutes. Titrate the liberated iodine with approximately 0.1 normal sodium thiosulfate solution (see 4.7.3.1). When the iodine color is nearly discharged add 5 ml. starch indicator solution (see 4.7.3.2) and continue the titration until the blue color just disappears. (1 ml. 0.1 N) sodium thiosulfate solution is equivalent to 0.00497 gm. of the dihydrate.) Calculate the percent, by weight, of sodium dichromate in the dried sample.

4.7.3.1 Standard sodium thiosulfate solution - Dissolve 25 grams reagent grade sodium thiosulfate crystals ($Na_2S_2O_3 \cdot 5H_2O$) and 0.2 gram sodium carbonate in a liter of freshly boiled and cooled distilled water. Following the procedure given in 4.7.3. Standarize the solution against 0.2 gram portions of recrystallized reagent grade potassium dichromate which has been dried to constant weight, or against a standard solution of this reagent.

4.7.3.2 Starch indicator solution - Stir up 5 grams of soluble starch with 100 ml. 1 percent salicylic acid solution, add 300 to 400 ml. boiling water and boil until the starch dissolves, Dilute to 1 liter.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Unless otherwise specified (see 6.2), the inhibitor shall be packaged in 1-quart, 1-gallon or 5-gallon containers. The metal cans shall conform to the requirements of Specification PPP-C-96, Type V, class 4.

5.2 Packing - Unless otherwise specified, unit packages of the inhibitor shall be packed in substantial commercial metal containers of the same size to insure acceptance and the delivery by common or other carriers at the lowest rate, to the point of delivery.

5.3 MARKING -

5.3.1 Individual containers.- Each individual container for the inhibitor shall be legibly labeled. The labels shall be printed with waterproof ink, except that any other suitable method of production.

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(such as mimeographing, multilithing, or lithographing) may be used provided that the labels so reproduced are made sufficiently waterproof or water repellent that legibility of the wording thereon shall not be reduced nor affected for a period of two years under normal conditions of storage and handling aboard ship, where continuous exposure to damp, salt air may reasonably be expected. The label shall contain a statement of intended use and instruction for using as follows:

COOLING SYSTEM CORROSION INHIBITOR

A DICHROMATE TYPE CORROSION INHIBITOR
FOR THE FRESH WATER COOLING SYSTEMS
OF INTERNAL COMBUSTION ENGINES

Initial and weekly doses should be added in accordance with the following:

Capacity Cooling System	Initial Dose	Weekly Dose
25 gal	1/2 pt (8 oz)	1 oz
50 gal	1 pt	2 oz
100 gal	2 pts	4 oz

5.3.2 The manufacturer's label may be used provided that all of the following information is included thereon, or a supplementary label be provided to include all points not covered by the manufacturer's lable. Such information shall include:

1. Stock number
2. Name of the material as given in the title of this specification
3. Specification number
4. Quantity in container
5. Contract number
6. Name of manufacturer
7. Manufacturer's lot number
8. Gross weight
9. A statement of intended use and the manner and limitations involved in such use of the inhibitor

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5.3.3 Shipping containers.- Unless otherwise specified, shipping containers shall be marked with the name of the material, quantity and size of containers therein, the stock number, the name of the contractor, the number of the contract or order, and the gross weight.

6. NOTES

6.1 Intended use.- The inhibitor is used to retard chemical reactions in closed cooling water systems of internal combustion engines.

6.2 Ordering data.- Procurement documents should stipulate the following:

(a) Title, number, and date of this specification.

(b) Size of containers (see 5.1).

(c) Federal stock numbers.

(d) Inspection procedure (see 4.4).

6.3 Bid samples. -. Unless otherwise specified, samples with bids are not required.

6.4 This document covers the requirements under MIL-I-19528A(CG)

NOTICE. - When Government drawings, specifications, or other data are used for any other purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity:
U. S. Coast Guard

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

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DEPARTMENT OF THE NAVY

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Commandant (G-FSL-3)
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Washington, D. C. 20590

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PLATE NO. 15419 (BACK)

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